

MATH 350 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 2
 - Types of real numbers
 - The tenth axiom

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The problem with axioms

Problems when axiomatizing:

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- intuitive axioms

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We need to redetermine basics

For us, we've lost some basic important things...

- We have 0 and 1 **by axioms**

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- what is 2?

Definition: $2 = 1 + 1$

- what is 3?

Definition: $3 = 2 + 1$

- are 1, 2, 3 all different?

Challenge!

Problem

Prove that 1, 2, and 3 are all different.

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Solution

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Since $0 < 1$, A7 says $0 + 1 < 1 + 1$.

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Since $0 < 1$, A7 says $0 + 1 < 1 + 1$.

Therefore $1 < 2$.

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Then again by A7, $1 + 1 < 2 + 1$.

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Also transitivity (A9) says $1 < 2$ and $2 < 3$ implies $1 < 3$.

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Since $0 < 1$, A7 says $0 + 1 < 1 + 1$.

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Then again by A7, $1 + 1 < 2 + 1$.

Therefore $2 < 3$.

Also transitivity (A9) says $1 < 2$ and $2 < 3$ implies $1 < 3$.

The trichotomy then implies $1 \neq 2$ and $2 \neq 3$ and $1 \neq 3$.

Natural numbers

Question

What are the natural numbers $1, 2, 3, \dots$ based on the Axioms?

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What are the natural numbers $1, 2, 3, \dots$ based on the Axioms?

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- a set $\mathbb{N}$
- $1 \in \mathbb{N}$
- $2 \in \mathbb{N}$
- $3 \in \mathbb{N}$
- general property:

$$x \in \mathbb{N} \implies x + 1 \in \mathbb{N}.$$

Inductive sets

Definition

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Is there more than one inductive set?

$$\mathbb{R}, \quad \mathbb{Q}, \quad (0, \infty), \quad \mathbb{Z}, \quad \mathbb{N}, \dots$$

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- integers are

$$\mathbb{Z} = \{x : x \text{ is zero or } \pm x \text{ is a positive integer}\}$$

- rational numbers are

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\}.$$

Induction

Principle of Induction:

If S is an inductive set, then $\mathbb{N} \subseteq S$

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Proof:

Let

$$S = \{n \in \mathbb{N} : 2|n(n+1)\}$$

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. We see that $1 \in S$ because 2 divides $1(1+1)$.

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Now suppose $x \in S$. (This is our usual inductive assumption).

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Proof continued:

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Example: Let's prove $n(n+1)$ is divisible by 2 for all positive integers n , using the Principle of Induction.

Proof continued:

Then 2 divides $x(x+1)$, so $x(x+1) = 2k$ for some integer k .

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This means $(x+1)(x+2) = x(x+1) + 2(x+1) = 2(k+x+1)$.

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Thus 2 divides $(x+1)(x+2)$, showing that $x+1 \in S$.

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By the Principle of Induction, this means $\mathbb{N} \subseteq S$.
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By the Principle of Induction, this means $\mathbb{N} \subseteq S$.
In other words $n \in S$ for every positive integer n .
Hence 2 divides $n(n+1)$ for every positive integers n .

Other numbers

Question

What about other kinds of numbers, like irrational algebraic numbers or transcendentals? Why do they exist **according to the axioms**?

To answer this, we need to understand the tenth axiom of the real numbers:

the completeness axiom

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Real numbers

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- **completeness axiom:** every nonempty set of numbers $S \subseteq F$ which is bounded above has a supremum $\sup(S)$
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- **completeness axiom:** every nonempty set of numbers $S \subseteq F$ which is bounded above has a supremum $\sup(S)$
- Understanding what this means is challenging, but will be fundamental to the entire course!
- Intuitively, it represents the fact that the real line has no holes or gaps.

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Examples:

- 34 is an upper bound of $[-1, 5]$, but not a maximal element
- 5 is a maximal element of $[-1, 5]$
- 3 is an upper bound of $[0, 3)$, but not a maximal element
- $\mathbb{N}$ has no upper bound
- $[3, 7)$ has an upper bound but no maximal element

Challenge!

Problem

Show that if S has a maximal element, then it is unique.

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Hint: use the definition!

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Solution

Suppose that b_1 and b_2 are both maximal elements of S .

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Solution

Suppose that b_1 and b_2 are both maximal elements of S .
Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.

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Solution

Suppose that b_1 and b_2 are both maximal elements of S .

Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.

Moreover, $b_1 \in S$ and $b_2 \in S$.

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Show that if S has a maximal element, then it is unique.

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Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.

Moreover, $b_1 \in S$ and $b_2 \in S$.

Since $b_1 \in S$ and b_2 is an upper bound of S , $b_1 \leq b_2$.

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Since $b_1 \in S$ and b_2 is an upper bound of S , $b_1 \leq b_2$.

Likewise, since $b_2 \in S$ and b_1 is an upper bound of S , $b_2 \leq b_1$.

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By the trichotomy, we find $b_1 = b_2$.

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Solution

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Then $x \leq b_1$ and $x \leq b_2$ for all $x \in S$.

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Now we can say *the* maximum, $\max(S)$

Challenge

Problem

Prove that if A is a nonempty set of integers that is bounded above, then A has a maximum.

Challenge

Solution

The set A is bounded above, so it has a supremum $b = \sup(A)$.

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The set A is bounded above, so it has a supremum $b = \sup(A)$. Since $b - 1$ is not an upper bound, so there exists $a \in A$ with $b - 1 < a$.

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If $a' \in A$, then a' is an integer and therefore $a' = a + k$ for $k \in \mathbb{Z}$.

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If $a' \in A$, then a' is an integer and therefore $a' = a + k$ for $k \in \mathbb{Z}$. Since b is an upper bound, $b \geq a + k > b - 1 + k$, making $0 > k - 1$.

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Solution

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Therefore $k \leq 0$ and $a' \leq a$.

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Solution

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If $a' \in A$, then a' is an integer and therefore $a' = a + k$ for $k \in \mathbb{Z}$. Since b is an upper bound, $b \geq a + k > b - 1 + k$, making $0 > k - 1$.

Therefore $k \leq 0$ and $a' \leq a$.

It follows that a is an upper bound of A , and since $a \in A$ it is a maximum.

Supremum

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In other words

a supremum is a **least upper bound**

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Solution

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Solution

Suppose that b_1 and b_2 are both suprema of S .
Then b_1 and b_2 are both upper bounds of S .

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Solution

Suppose that b_1 and b_2 are both suprema of S .
Then b_1 and b_2 are both upper bounds of S .
Since b_1 is a least upper bound, $b_1 \leq b_2$.

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Suppose that b_1 and b_2 are both suprema of S .

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Since b_2 is also a least upper bound, $b_2 \leq b_1$.

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Show that if S has a maximum element. Then S has a supremum and $\max(S) = \sup(S)$

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Solution

Let $b = \max(S)$.

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Solution

Let $b = \max(S)$.

Then b is an upper bound of S and $b \in S$.

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If b' is another upper bound of S , then by definition $b \leq b'$.

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Solution

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Thus b is the least upper bound of S .

Completeness Axiom

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- 2 is the supremum of

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- 1 is the supremum of

$$\{\sin(n) : n \in \mathbb{N}\}$$

Decimals

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The k 'th **decimal apprixomation** of $n.d_1d_2d_3\dots$ is

$$s_k = n + \frac{d_1}{10} + \frac{d_2}{10^2} + \frac{d_3}{10^3} + \dots + \frac{d_k}{10^k}.$$

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The **decimal** $n.d_0d_1d_2d_3\dots$ is the real number defined as the supremum of decimal approximations

$$n.d_0d_1d_2d_3\dots = \sup\{s_k : k \in \mathbb{N} \dots\}.$$

Examples of decimals

- Liouville's constant

$$0.11000100000000000000000010\dots$$

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- later we will show *any* positive number has at least one decimal expansion

Lower bounds and infima

A **lower bound** for a set $S \subseteq \mathbb{R}$ is a number b such that

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In other words

an infimum is a **greatest lower bound**

Properties of Suprema

The first property of suprema is that they must be arbitrarily close to elements of the set.

Theorem (Approximation Property)

Let $S \subseteq \mathbb{R}$ be nonempty and bounded above, and let $b = \sup(S)$. Then for all $\epsilon > 0$, there exists $a \in S$ with

$$b - \epsilon < a \leq b.$$

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Therefore there must exist $a \in S$ with $a > b - \epsilon$. □

Properties of Suprema

Also, suprema play nicely with addition.

Theorem (Additive Property)

Let $A, B \subseteq \mathbb{R}$ be nonempty, bounded above sets.

$$C = \{x + y : x \in A, y \in B\}.$$

Then C is bounded above and

$$\sup(C) = \sup(A) + \sup(B).$$

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Therefore $a + b \leq c$. □

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It follows from Axiom 7 that $b < n + 1$.

However, $n + 1 \in \mathbb{Z}$, so this contradicts b being an upper bound.

Archimedean property

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For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{N}$ with $y < nx$.

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Proof.

Replace x with y/x in the previous theorem. □